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SUBSTRATE FIXING DEVICE

Abstract

A substrate fixing device includes a ceramic base plate, a ceramic electrostatic chuck having a mounting surface on which an object to be sucked is mounted, and a first brazing part joining the base plate and the electrostatic chuck. The base plate is made of a ceramic material having a thermal expansion coefficient of 0 ± 3 ppm/K at a room temperature. The electrostatic chuck is made of a ceramic material having a thermal expansion coefficient of 0 ± 3 ppm/K at the room temperature.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a substrate fixing device.

BACKGROUND ART

[0002] In the related art, a film forming device (for example, a chemical vapor deposition (CVD) device or a physical vapor deposition (PVD) device) or a plasma etching device used for manufacturing a semiconductor element such as an IC or an LSI has a substrate fixing device for accurately holding a substrate such as a silicon wafer in a vacuum processing chamber. In a substrate fixing device, a ceramic electrostatic chuck is bonded onto a metal base plate by an adhesive layer (for example, see JP2020-23088A).

SUMMARY OF INVENTION

[0003] In the substrate fixing device of the related art, a temperature of a mounting surface of the electrostatic chuck on which an object to be sucked is mounted may vary. Specifically, when the substrate fixing device is exposed to a low temperature of about -60°C . or a high temperature of about 180°C ., a large difference occurs between a thermal deformation amount of the electrostatic chuck and a thermal deformation amount of the base plate.

[0004] When a large stress acts on the adhesive layer, cohesive failure may occur in the adhesive layer. When cohesive failure occurs in the adhesive layer, in-plane uniformity of a thermal resistance of the adhesive layer is reduced, so that the temperature of the mounting surface of the electrostatic chuck varies.

[0005] According to one aspect of the present disclosure, there is provided a substrate fixing device includes a ceramic base plate, a ceramic electrostatic chuck having a mounting surface on which an object to be sucked is mounted, and a first brazing part joining the base plate and the electrostatic chuck. The base plate is made of a ceramic material having a thermal expansion coefficient of 0 ± 3 ppm/K at a room temperature. The electrostatic chuck is made of a ceramic material having a thermal expansion coefficient of 0 ± 3 ppm/K at the room temperature.

[0006] According to one aspect of the present disclosure, it is possible to improve the uniformity of the temperature of the mounting surface.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1 is a schematic cross-sectional view showing a substrate fixing device according to an embodiment.

[0008] FIG. 2 is an enlarged cross-sectional view showing a part of the substrate fixing device shown in FIG. 1.

[0009] FIG. 3 is a schematic cross-sectional view showing a manufacturing method for the substrate fixing device.

[0010] FIG. 4 is a schematic cross-sectional view showing the manufacturing method for the substrate fixing device.

[0011] FIG. 5 is a schematic cross-sectional view showing the manufacturing method for the substrate fixing device.

[0012] FIG. 6 is a schematic cross-sectional view showing a substrate fixing device according to a modification.

DESCRIPTION OF EMBODIMENTS

[0013] Hereinafter, an embodiment will be described with reference to the accompanying drawings.

[0014] In addition, in the accompanying drawings, for the sake of convenience, a portion serving as characteristics may be shown in an enlarged manner in order to facilitate understanding of the characteristics, and a dimensional ratio of each component may be different in each drawing. In the

cross-sectional view, hatching of some members is shown instead of a satin-like pattern in order to facilitate understanding of a cross-sectional structure of each member, and hatching of some members is omitted. In this specification, an “upper-lower direction” and a “left-right direction” are directions in a case where a direction in which a sign indicating each member is correctly read in each drawing is a positive position.

Overall Configuration of Substrate Fixing Device 10

[0015] As shown in FIG. 1, the substrate fixing device 10 includes a ceramic base plate 20, a ceramic electrostatic chuck 70, and a brazing part 80 joining the base plate 20 and the electrostatic chuck 70. The substrate fixing device 10 includes, for example, a protective layer 90 that protects the brazing part 80. The electrostatic chuck 70 is fixed to an upper surface of the base plate 20 by the brazing part 80. The substrate fixing device 10 is a device that sucks and holds a substrate (not shown) that is an object to be sucked by the electrostatic chuck 70 mounted on the upper surface of the base plate 20. Examples of the substrate include a silicon wafer. A diameter of the substrate may be, for example, about 8 inches, 12 inches, or 18 inches.

[0016] A low thermal expansion ceramic material having a low thermal expansion coefficient can be used as materials of the base plate 20 and the electrostatic chuck 70. As the low thermal expansion ceramic material, for example, a ceramic material having a thermal expansion coefficient of 0 ± 3 ppm/K at a room temperature can be used. If the thermal expansion coefficient is 0 ± 3 ppm/K, cohesive failure of the adhesive layer can be prevented. In addition, as a low thermal expansion ceramic material, it is more preferable to use a ceramic material having a thermal expansion coefficient of 0 ± 1 ppm/K at the room temperature. That is, a ceramic material having a thermal expansion coefficient of zero can be used as the low thermal expansion ceramic material. As the low thermal expansion ceramic material, for example, a ceramic material having cordierite ($2\text{MgO} \cdot \text{Math.}2\text{Al} \cdot \text{sub.}2\text{O} \cdot \text{sub.}3 \cdot \text{Math.}5\text{SiO} \cdot \text{sub.}2$) as a main component can be used. Here, the “main component” in the specification refers to a component that occupies 90% by weight or more of the components contained in a target site. The material of the base plate 20 and the material of the electrostatic chuck 70 may be the same ceramic material or different ceramic materials. The room temperature is a temperature range of about 22° C. to 26° C.

Configuration of Base Plate 20

[0017] The base plate 20 is a base (base) on which the electrostatic chuck 70 is mounted. The base plate 20 has rigidity for supporting the electrostatic chuck 70. A thickness of the base plate 20 can be, for example, approximately 20 mm to 50 mm.

[0018] The base plate 20 includes a lower portion 21 and an upper portion 22 laminated on an upper surface of the lower portion 21. The lower portion 21 is formed in a disk shape, for example. The upper portion 22 is formed in a disk shape, for example. For example, the upper portion 22 is concentrically disposed on the upper surface of the lower portion 21. A planar size of the upper portion 22 is smaller than a planar size of the lower portion 21. It should be noted that a “planar size” in this specification is a size in a direction perpendicular to a laminating direction of the base plate 20, the brazing part 80, and the electrostatic chuck 70, that is, in the left-right direction. A diameter of the upper portion 22 is smaller than a diameter of the lower portion 21. The upper portion 22 protrudes upward from the upper surface of the lower portion 21.

[0019] A cooling path 30 is provided inside the base plate 20, for example. The cooling path 30 includes an introduction portion 31 provided at one end thereof and a discharge portion 32 provided at the other end thereof. The cooling path 30 is, for example, connected to a cooling medium control device (not shown) provided outside the substrate fixing device 10. The cooling medium control device introduces the cooling medium from the introduction portion 31 to the cooling path 30 and discharges the cooling medium from the discharge portion 32. By circulating a cooling medium through the cooling path 30 to cool the base plate 20, the substrate sucked to the electrostatic chuck 70 can be cooled. For example, water or Galden may be used as the cooling medium.

[0020] A gas flow path **40**, for example, is provided inside the base plate **20**. The gas flow path **40** penetrates the base plate **20** in a thickness direction (an upper-lower direction in the drawing). Specifically, the gas flow path **40** penetrates from the upper surface of the upper portion **22** to a lower surface of the lower portion **21**. For example, a gas for cooling the substrate sucked on the electrostatic chuck **70** is introduced into the gas flow path **40**. An inert gas may be used as gas for cooling. Examples of the inert gas include helium (He) gas and argon (Ar) gas.

[0021] The base plate **20** has, for example, a structure in which a plurality of (here, two layers) ceramic plates **51** and **52** are laminated. The base plate **20** includes the ceramic plates **51** and **52** and a brazing part **60** joining the ceramic plates **51** and **52**. The ceramic plate **52** is joined to the upper surface of the ceramic plate **51** by the brazing part **60**.

[0022] The ceramic plate **51** constitutes the lower portion **21** of the base plate **20**, for example. The ceramic plate **51** is formed in a disk shape, for example. The ceramic plate **51** includes, for example, the introduction portion **31**, the discharge portion **32**, and recessed portions **33** that constitute the cooling path **30**.

[0023] The recessed portion **33** is formed to be recessed downward from the upper surface of the ceramic plate **51**. The recessed portion **33** is formed to open upward in the ceramic plate **51**. The introduction portion **31** is formed to be recessed upward from the lower surface of the ceramic plate **51** and to communicate with the recessed portion **33**. The introduction portion **31** is formed to open downward in the ceramic plate **51**. The discharge portion **32** is formed to be recessed upward from the lower surface of the ceramic plate **51** and to communicate with the recessed portion **33**. The discharge portion **32** is formed to open downward in the ceramic plate **51**.

[0024] As shown in FIG. 2, the ceramic plate **51** has a hole portion **41** constituting the gas flow path **40**. The hole portion **41** penetrates the ceramic plate **51** in the thickness direction. The hole portion **41** is formed to open upward in the ceramic plate **51** and also to open downward in the ceramic plate **51**. The hole portion **41** is formed to extend linearly along the thickness direction of the ceramic plate **51**, for example.

[0025] The ceramic plate **52** constitutes, for example, the upper portion **22** of the base plate **20**. The ceramic plate **52** is formed in a disk shape, for example. A planar size of the ceramic plate **52** is smaller than a planar size of the ceramic plate **51**.

[0026] The ceramic plate **52** is provided to close an opening of the recessed portion **33** of the ceramic plate **51**. Thus, the cooling path **30** is formed by the ceramic plate **52** that closes the opening of the recessed portion **33**, the recessed portion **33**, the introduction portion **31**, and the discharge portion **32** (see FIG. 1).

[0027] The ceramic plate **52** has hole portions **42** and **43** constituting the gas flow path **40**. The hole portion **42** and the hole portion **43** communicate with each other. The hole portion **42** and the hole portion **43** are formed to penetrate the ceramic plate **52** in the thickness direction in cooperation with each other. The hole portion **42** is formed to be recessed upward from a lower surface of the ceramic plate **52**. The hole portion **42** is formed to open downward in the ceramic plate **52**. The hole portion **42** communicates with the hole portion **41** of the ceramic plate **51**. The hole portion **42** is formed to extend linearly along the thickness direction of the ceramic plate **52**, for example.

[0028] The hole portion **43** is formed to be recessed downward from the upper surface of the ceramic plate **52**, and is formed to communicate with the hole portion **42**. The hole portion **43** is formed to open upward in the ceramic plate **52**. A planar size of the hole portion **43** is, for example, larger than a planar size of the hole portion **42**. That is, an opening area of the hole portion **43** is larger than an opening area of the hole portion **42**. The hole portion **43** is provided so as to overlap the entire hole portion **42** in a planar view. Further, the ceramic plate **52** may be provided with a groove portion formed to be recessed downward from the upper surface of the ceramic plate **52**.

[0029] The brazing part **60** includes a conductive pattern **61** formed on the upper surface of the ceramic plate **51**, a conductive pattern **62** formed on the lower surface of the ceramic plate **52**, and a brazing member **63** joining the conductive patterns **61** and **62**.

[0030] For example, the conductive pattern **61** is provided so as to overlap the ceramic plate **52** in a planar view. For example, the conductive pattern **61** is provided so as not to overlap the recessed portion **33** and the hole portion **41** in a planar view. For example, the conductive pattern **62** is provided so as to overlap the conductive pattern **61** in a planar view. For example, the conductive pattern **62** is provided so as not to overlap the recessed portion **33** and the hole portions **41** and **42** in a planar view.

[0031] The brazing member **63** joins the conductive pattern **61** and the conductive pattern **62**. The brazing member **63** is joined to an upper surface of the conductive pattern **61** and also joined to a lower surface of the conductive pattern **62**. Thus, the conductive pattern **61** and the conductive pattern **62** are joined to each other by the brazing using the brazing member **63**. The ceramic plate **51** and the ceramic plate **52** are joined to each other by joining the conductive pattern **61** and the conductive pattern **62** to each other. Thus, the ceramic plate **52** is laminated on the upper surface of the ceramic plate **51**. As the brazing member **63**, for example, silver brazing can be used.

Configuration of Electrostatic Chuck **70**

[0032] As shown in FIG. **1**, the electrostatic chuck **70** includes a substrate body **71** and an electrode **72** embedded in the substrate body **71**. The electrostatic chuck **70** is, for example, a Johnsen-Rahbek type electrostatic chuck. However, the electrostatic chuck **70** may be a Coulomb force type electrostatic chuck. The electrostatic chuck **70** is a holder that sucks and holds a substrate which is an object to be sucked.

[0033] The substrate body **71** is formed in a disk shape, for example. For example, a diameter of the substrate body **71** may be equal to the diameter of the upper portion **22** of the base plate **20** or may be larger than the diameter of the upper portion **22**. The diameter of the substrate body **71** according to the present embodiment is equal to the diameter of the upper portion **22**. The diameter of the substrate body **71** may be, for example, about 150 mm to 500 mm. A thickness of the substrate body **71** may be, for example, about 0.5 mm to 10 mm.

[0034] The substrate body **71** has a mounting surface **71A** (here, an upper surface) on which a substrate as an object to be sucked is mounted. The substrate body **71** is, for example, a dielectric. The substrate body **71** is, for example, a ceramic substrate.

[0035] The substrate body **71** is provided with, for example, a gas flow path **73**. The gas flow path **73** penetrates the substrate body **71** in the thickness direction. The gas flow path **73** is formed to open upward in the substrate body **71**, and also to open downward in the substrate body **71**. The gas flow path **73** communicates with the gas flow path **40** of the base plate **20**. As shown in FIG. **2**, the gas flow path **73** communicates with the hole portion **43** of the ceramic plate **52**. The gas flow path **73** is formed to extend linearly along the thickness direction of the substrate body **71**, for example. A planar size of the gas flow path **73** is, for example, smaller than the planar size of the hole portion **43**, for example. That is, an opening area of the gas flow path **73** is smaller than the opening area of the hole portion **43**. The gas flow path **73** as a whole is provided so as to overlap the hole portion **43** in a planar view. For example, the gas flow path **73** is provided at a position different from that of the hole portion **42** in a planar view.

[0036] As shown in FIG. **1**, in the substrate fixing device **10**, a gas hole **11** is formed by the gas flow path **40** and the gas flow path **73**. The gas hole **11** penetrates from the mounting surface **71A** of the substrate body **71** to the lower surface of the base plate **20** by the communication between the gas flow path **40** and the gas flow path **73**. In the gas hole **11**, the inert gas is introduced into the gas hole **11** through the gas flow path **40**, and the inert gas is discharged from the gas hole **11** through the gas flow path **73**. The inert gas discharged from the gas flow path **73** is, for example, filled between the lower surface of the substrate placed on the mounting surface **71A** and the mounting surface **71A**, so that the substrate can be cooled.

[0037] The electrode **72** is, for example, an electrostatic electrode for sucking a substrate placed on the mounting surface **71A**. The electrode **72** is an electrode formed in a thin film shape. The electrode **72** is built in, for example, a portion of the substrate body **71** located near the mounting

surface **71A** in the thickness direction. For example, the electrode **72** is disposed on a plane parallel to the mounting surface **71A**. The electrode **72** is electrically connected to, for example, a power supply for suction provided outside the substrate fixing device **10**. When a predetermined voltage is applied from the power supply for suction, the electrode **72** generates an electrostatic suction force between the electrode **72** and the substrate placed on the mounting surface **71A**. Accordingly, the substrate can be sucked and held on the mounting surface **71A**. A suction holding force of the electrostatic chuck **70** increases as the voltage applied to the electrode **72** increases. The electrode **72** may have a unipolar shape or a bipolar shape. As a material of the electrode **72**, for example, tungsten (W) or molybdenum (Mo) can be used. In each drawing, one electrode **72** is shown, but actually, a plurality of electrodes arranged on the same plane are included.

Configuration of Brazing Part **80**

[0038] As shown in FIG. **2**, the brazing part **80** includes a conductive pattern **81** formed on the upper surface of the base plate **20**, a conductive pattern **82** formed on the lower surface of the electrostatic chuck **70**, and a brazing member **83** joining the conductive patterns **81** and **82**.

[0039] The conductive pattern **81** is formed on the upper surface of the ceramic plate **52**. For example, the conductive pattern **81** is provided so as to overlap the electrostatic chuck **70** in a planar view. For example, the conductive pattern **81** is provided so as not to overlap the hole portion **43** and the gas flow path **73** in a planar view. For example, the conductive pattern **81** covers the entire upper surface of the ceramic plate **52**.

[0040] The conductive pattern **82** is formed on a lower surface of the substrate body **71**. For example, the conductive pattern **82** is provided so as to overlap the conductive pattern **81** in a planar view. For example, the conductive pattern **82** is provided so as not to overlap the hole portion **43** and the gas flow path **73** in a planar view.

[0041] The brazing member **83** joins the conductive pattern **81** and the conductive pattern **82**. The brazing member **83** is joined to the upper surface of the conductive pattern **81** and also joined to the lower surface of the conductive pattern **82**. Thus, the conductive pattern **81** and the conductive pattern **82** are joined to each other by the brazing using the brazing member **83**. The base plate **20** and the electrostatic chuck **70** are joined to each other by joining the conductive pattern **81** and the conductive pattern **82** to each other. Thus, the electrostatic chuck **70** is laminated on the upper surface of the base plate **20**. As the brazing member **83**, for example, silver brazing can be used.

Configuration of Protective Layer **90**

[0042] The protective layer **90** has a function of protecting the brazing parts **60** and **80** from plasma. The protective layer **90** is formed on an outer side surface of the base plate **20** and an outer side surface of the electrostatic chuck **70** so as to cover the brazing parts **60** and **80**. The protective layer **90** covers the entire outer side surface of the brazing part **60**. That is, the protective layer **90** covers the entire outer side surface of the conductive pattern **61**, the entire outer side surface of the conductive pattern **62**, and the entire outer side surface of the brazing member **63**. The protective layer **90** covers the entire outer side surface of the brazing part **80**. That is, the protective layer **90** covers the entire outer side surface of the conductive pattern **81**, the entire outer side surface of the conductive pattern **82**, and the entire outer side surface of the brazing member **83**. The protective layer **90** covers the outer side surfaces of the brazing parts **60** and **80** over the entire circumference of the base plate **20**. The protective layer **90** continuously covers the brazing part **60** and the brazing part **80**, for example.

[0043] The protective layer **90** covers, for example, the entire outer side surface of the upper portion **22** of the base plate **20**. That is, the protective layer **90** covers the entire outer side surface of the ceramic plate **52**. The protective layer **90** covers, for example, a part of the outer side surface of the electrostatic chuck **70**. The protective layer **90** covers, for example, a part of the outer side surface of the substrate body **71**. The protective layer **90** of the present embodiment covers the outer side surface in the lower portion of the substrate body **71** and to expose the outer side surface in the upper portion of the substrate body **71**.

[0044] As a material of the protective layer **90**, for example, a low thermal expansion ceramic material can be used. Examples of the material of the protective layer **90** include ceramic materials such as aluminum oxide, aluminum nitride, and yttrium oxide. As the material of the protective layer **90**, for example, a fluororesin or an epoxy resin having excellent plasma resistance may be used.

[0045] In the present embodiment, the gas flow path **40** is an example of a first gas flow path, the hole portion **41** is an example of a first hole portion, the hole portions **42** and **43** are examples of a second hole portion, and the gas flow path **73** is an example of a second gas flow path. The ceramic plate **51** is an example of a first ceramic plate, and the ceramic plate **52** is an example of a second ceramic plate. The brazing part **60** is an example of a second brazing part, the conductive pattern **61** is an example of a third conductive pattern, the conductive pattern **62** is an example of a fourth conductive pattern, and the brazing member **63** is an example of a second brazing member. The brazing part **80** is an example of a first brazing part, the conductive pattern **81** is an example of a first conductive pattern, the conductive pattern **82** is an example of a second conductive pattern, and the brazing member **83** is an example of a first brazing member.

Manufacturing Method for Substrate Fixing Device **10**

[0046] Next, a manufacturing method for the substrate fixing device **10** will be described. For convenience of description, the parts that become components of the substrate fixing device **10** will be described with reference numerals of the final components.

[0047] First, in a step shown in FIG. **3**, the ceramic plate **51** whose upper surface is formed with the conductive pattern **61** for brazing is prepared. In addition, the ceramic plate **52** whose lower surface is formed with the conductive pattern **62** for brazing and whose upper surface is formed with the conductive pattern **81** for brazing is prepared. In addition, the electrostatic chuck **70** having the substrate body **71** whose lower surface is formed with the conductive pattern **82** for brazing is prepared. The two ceramic plates **51** and **52** and the substrate body **71** can be manufactured by, for example, a green sheet method.

[0048] Subsequently, in a step shown in FIG. **4**, the introduction portion **31**, the discharge portion **32** (see FIG. **1**), and the recessed portion **33** are formed in the ceramic plate **51**, and the hole portion **41** is formed in the ceramic plate **51**. The introduction portion **31**, the discharge portion **32**, the recessed portion **33**, and the hole portion **41** can be formed by, for example, a laser processing method or a machining processing method.

[0049] In the step shown in FIG. **4**, the hole portion **42** and the hole portion **43** are formed in the ceramic plate **52**. The hole portion **42** and the hole portion **43** can be formed by, for example, a laser processing method or a machining processing method.

[0050] In the step shown in FIG. **4**, the gas flow path **73** is formed in the substrate body **71**. The gas flow path **73** can be formed by, for example, a laser processing method or a machining processing method.

[0051] Next, the ceramic plate **52** is disposed above the ceramic plate **51** in a state in which the conductive pattern **62** faces the conductive pattern **61**. At this time, the ceramic plates **51** and **52** are aligned such that the hole portion **42** overlap the hole portion **41** in a planar view. The substrate body **71** is disposed above the ceramic plate **52** in a state in which the conductive pattern **82** faces the conductive pattern **81**. At this time, the ceramic plate **52** and the substrate body **71** are aligned such that the gas flow path **73** overlaps the hole portion **43** in a planar view.

[0052] Next, in a step shown in FIG. **5**, the conductive pattern **61** and the conductive pattern **62** are joined by the brazing using the brazing member **63**. Accordingly, the ceramic plate **52** can be joined to the upper surface of the ceramic plate **51**. Further, the conductive pattern **81** and the conductive pattern **82** are joined by the brazing using the brazing member **83**. Accordingly, the substrate body **71** can be joined to the upper surface of the ceramic plate **52**. According to this step, the opening of the recessed portion **33** is closed by the ceramic plate **52**. As a result, the cooling path **30** is formed by the introduction portion **31**, the discharge portion **32** (see FIG. **1**), the recessed portion **33**

formed in the ceramic plate **51**, and the ceramic plate **52** that closes the opening of the recessed portion **33**. Further, according to the present step, the gas flow path **40** is formed by communicating the hole portion **41** of the ceramic plate **51** with the hole portions **42** and **43** of the ceramic plate **52**, and the gas flow path **40** communicates with the gas flow path **73** to form the gas hole **11**.

[0053] According to the manufacturing steps described above, the base plate **20** having the ceramic plates **51** and **52** is formed, and the electrostatic chuck **70** is joined onto the base plate **20**.

[0054] Next, the protective layer **90** that covers the brazing parts **60** and **80** is formed on the outer side surface of the ceramic plate **52** and the outer side surface of the substrate body **71**. The protective layer **90** can be formed by, for example, a physical vapor deposition method or a chemical vapor deposition method. When a fluororesin or an epoxy resin is used as the material of the protective layer **90**, for example, the protective layer **90** can be formed by applying a liquid resin to the outer side surface of the ceramic plate **52** and the outer side surface of the substrate body **71** and then curing the resin.

[0055] According to the above manufacturing steps, the substrate fixing device **10** according to the present embodiment can be manufactured.

[0056] Next, the operation and effect of the present embodiment will be described.

[0057] (1) The substrate fixing device **10** includes the ceramic base plate **20**, the ceramic electrostatic chuck **70** having the mounting surface **71A** on which an object to be sucked is mounted, and the brazing part **80** joining the base plate **20** and the electrostatic chuck **70**. The base plate **20** is made of a ceramic material having a thermal expansion coefficient of 0 ± 5 ppm/K at the room temperature. The electrostatic chuck **70** is made of a ceramic material having a thermal expansion coefficient of 0 ± 5 ppm/K at the room temperature.

[0058] According to this configuration, the base plate **20** and the electrostatic chuck **70** are both made of a low thermal expansion ceramic material. Therefore, even when the substrate fixing device **10** is exposed to a low temperature of about -60°C . or a high temperature of about 180°C ., thermal deformation, such as thermal expansion or thermal contraction, hardly occurs in the base plate **20** and the electrostatic chuck **70**. Accordingly, even when the substrate fixing device **10** is exposed to a low temperature of about -60°C . or a high temperature of about 180°C ., it is possible to suitably reduce a large difference between a thermal deformation amount of the base plate **20** and a thermal deformation amount of the electrostatic chuck **70**. As a result, it is possible to suitably prevent thermal stress caused by thermal expansion and thermal contraction from acting on the brazing part **80** provided between the base plate **20** and the electrostatic chuck **70**. As a result, it is possible to suitably prevent the brazing part **80** from being broken due to the thermal stress. Therefore, it is possible to suitably prevent the decrease in in-plane uniformity of the thermal resistance of the brazing part **80** due to the breakdown of the brazing part **80**. As a result, variation in the temperature of the mounting surface **71A** of the electrostatic chuck **70** can be prevented, and uniformity of the temperature of the mounting surface **71A** of the electrostatic chuck **70** can be improved.

[0059] (2) During an operation of the substrate fixing device **10**, the temperature of the base plate **20** and the temperature of the electrostatic chuck **70** may be different from each other. At this time, even when the base plate **20** and the electrostatic chuck **70** are made of the same ceramic material, when the thermal expansion coefficient of the ceramic material is 6 ppm/K to 8 ppm/K, the thermal stress is generated due to a difference in thermal expansion due to the temperature difference.

[0060] On the other hand, in the substrate fixing device **10** according to the present embodiment, the base plate **20** and the electrostatic chuck **70** are both made of a low thermal expansion ceramic material. Therefore, even when the temperature of the base plate **20** and the temperature of the electrostatic chuck **70** are different from each other, the thermal deformation hardly occurs in the base plate **20** and the electrostatic chuck **70**. Accordingly, it is possible to suitably reduce a large difference between the thermal deformation amount of the base plate **20** and the thermal deformation amount of the electrostatic chuck **70**. As a result, it is possible to suitably prevent the

thermal stress caused by the thermal expansion and the thermal contraction from acting on the brazing part **80** that joins the base plate **20** and the electrostatic chuck **70**.

[0061] (3) Even when the substrate fixing device **10** is exposed to a low temperature of about -60° C. or a high temperature of about 180° C., it is possible to prevent the substrate fixing device **10** (that is, the base plate **20**, the electrostatic chuck **70**, and the brazing part **80**) from being damaged. Therefore, the single substrate fixing device **10** can be suitably used in a wide temperature range.

[0062] (4) In the substrate fixing device **10**, the base plate **20** is more preferably made of a ceramic material having a thermal expansion coefficient of 0 ± 1 ppm/K at the room temperature. The electrostatic chuck **70** is more preferably made of a ceramic material having a thermal expansion coefficient of 0 ± 1 ppm/K at the room temperature. Accordingly, even when the temperature of the base plate **20** and the temperature of the electrostatic chuck **70** are different from each other, the occurrence of thermal deformation in the base plate **20** and the electrostatic chuck **70** is further prevented. Therefore, the difference between the thermal deformation amount of the base plate **20** and the thermal deformation amount of the electrostatic chuck **70** can be further suitably prevented. As a result, it is possible to more suitably prevent the thermal stress caused by thermal expansion and the thermal contraction from acting on the brazing part **80** that joins the base plate **20** and the electrostatic chuck **70**.

[0063] (5) The protective layer **90** is formed on the outer side surface of the base plate **20** and the outer side surface of the electrostatic chuck **70** so as to cover the brazing parts **60** and **80**. The protective layer **90** can protect the brazing parts **60** and **80** from plasma. As a result, it is possible to prevent the brazing parts **60** and **80** from being deteriorated by plasma, so that the lifetime of the substrate fixing device **10** can be prolonged.

[0064] (6) The brazing part **80** includes the conductive pattern **81** formed on the upper surface of the base plate **20**, the conductive pattern **82** formed on the lower surface of the electrostatic chuck **70**, and the brazing member **83**. In this configuration, the electrostatic chuck **70** is joined onto the base plate **20** by joining the conductive patterns **81** and **82** by the brazing using the brazing member **83**. The substrate fixing device **10** thus formed has a structure having the conductive patterns **81** and **82**. Therefore, the rigidity of the substrate fixing device **10** can be increased, for example, as compared with a case where the base plate **20** and the electrostatic chuck **70** are joined to each other by an adhesive.

[0065] (7) The base plate **20** is constituted by the plurality of ceramic plates **51** and **52** joined to each other by the brazing part **60**. In this configuration, the conductive patterns **61** and **62** are provided for the ceramic plates **51** and **52**. Then, the conductive patterns **61** and **62** are joined by brazing with the brazing member **63**, whereby the ceramic plate **52** is joined onto the ceramic plate **51**. The base plate **20** thus formed has a structure having the conductive patterns **61** and **62**. Therefore, the rigidity of the base plate **20** can be increased, for example, as compared with a case where the ceramic plates **51** and **52** are joined to each other by an adhesive.

[0066] (8) The plurality of ceramic plates **51** and **52** are joined to the brazing part **60**, and the base plate **20** and the electrostatic chuck **70** are joined to the brazing part **80**. According to this configuration, the ceramic plates **51** and **52** and the substrate body **71** can be joined to each other by the brazing members **63** and **83** that are harder than the adhesive. This can improve airtightness of the cooling path **30** formed in the ceramic plates **51** and **52** and the gas hole **11** formed in the substrate body **71**.

OTHER EMBODIMENTS

[0067] The above embodiment can be modified as follows. The above embodiment and the following modification can be combined with each other within a technical contradiction. [0068] The structure of the substrate fixing device **10** according to the above embodiment can be appropriately changed. [0069] For example, as shown in FIG. **6**, the protective layer **90** may be formed so as to cover the entire outer side surface of the upper portion **22** of the base plate **20** and to cover the entire outer side surface of the electrostatic chuck **70**. That is, the protective layer **90**

may be formed so as to cover the entire outer side surface of the ceramic plate **52** and the entire outer side surface of the substrate body **71**. [0070] In the modification shown in FIG. **6**, the protective layer **90** may cover the entire outer side surface of the lower portion **21** of the base plate **20**. That is, the protective layer **90** may cover the entire outer side surface of the ceramic plate **51**. [0071] The protective layer **90** according to the above embodiment may be omitted. [0072] The electrostatic chuck **70** according to the above embodiment may include an electrode different from the electrode **72**. Examples of the other electrode include a heating element for heating the substrate placed on the mounting surface **71A**. [0073] An embossing may be provided on the mounting surface **71A** of the electrostatic chuck **70** according to the above embodiment. [0074] A shape of the gas hole **11** in the substrate fixing device **10** according to the above embodiment may be appropriately changed. The gas hole **11** may be omitted. [0075] A shape of the cooling path **30** in the base plate **20** according to the above embodiment may be appropriately changed. The cooling path **30** may be omitted. [0076] In the above embodiment, the base plate **20** and the electrostatic chuck **70** are joined to each other by the brazing part **80**, but the present invention is not limited thereto. For example, the base plate **20** and the electrostatic chuck **70** may be joined together by joining the conductive patterns **81** and **82** to each other by a conductive adhesive. [0077] In the base plate **20** according to the above embodiment, the ceramic plates **51** and **52** are joined to each other by the brazing part **60**, but the present invention is not limited thereto. For example, the conductive patterns **61** and **62** may be joined to each other by a conductive adhesive. The conductive patterns **61** and **62** may be omitted, and the ceramic plates **51** and **52** may be joined to each other by an adhesive such as a silicone adhesive. [0078] The number of the ceramic plates **51** and **52** in the base plate **20** according to the above embodiment is not particularly limited. For example, the number of ceramic plates included in the base plate **20** may be one, or three or more. [0079] In the manufacturing method according to the above embodiment, the recessed portion **33** and the hole portions **41**, **42**, and **43** are formed on the ceramic plates **51** and **52** after the green sheet is fired, but the present invention is not limited thereto. For example, the recessed portion **33** and the hole portions **41**, **42**, and **43** may be formed on the green sheet before the green sheet is fired, that is, on the green sheet before firing. [0080] In the manufacturing method according to the above embodiment, the gas flow path **73** is formed on the substrate body **71** after the green sheet is fired, but the present invention is not limited thereto. For example, the gas flow path **73** may be formed on the green sheet before the green sheet is fired, that is, on the green sheet before firing. [0081] The substrate fixing device **10** according to the above embodiment is applied to a semiconductor manufacturing device, for example, a dry etching device. Examples of the dry etching device include a parallel plate type reactive ion etching device. The substrate fixing device **10** can also be applied to a semiconductor manufacturing device such as a plasma chemical vapor deposition (CVD) device or a sputtering device.

Claims

1. A substrate fixing device comprising: a ceramic base plate; a ceramic electrostatic chuck having a mounting surface on which an object to be sucked is mounted; and a first brazing part joining the base plate and the electrostatic chuck, wherein the base plate is made of a ceramic material having a thermal expansion coefficient of 0 ± 3 ppm/K at a room temperature, and the electrostatic chuck is made of a ceramic material having a thermal expansion coefficient of 0 ± 3 ppm/K at the room temperature.
2. The substrate fixing device according to claim 1, wherein the base plate is made of a ceramic material having a thermal expansion coefficient of 0 ± 1 ppm/K at the room temperature, and the electrostatic chuck is made of a ceramic material having a thermal expansion coefficient of 0 ± 1 ppm/K at the room temperature.
3. The substrate fixing device according to claim 1, further comprising: a protective layer formed

on an outer side surface of the base plate and an outer side surface of the electrostatic chuck so as to cover the first brazing part.

4. The substrate fixing device according to claim 3, wherein the base plate has a lower portion and an upper portion laminated on an upper surface of the lower portion, a planar size of the upper portion is formed smaller than a planar size of the lower portion, and the protective layer is formed so as to cover an entire outer side surface of the upper portion and to cover an entire outer side surface of the electrostatic chuck.

5. The substrate fixing device according to claim 1, wherein the first brazing part includes a first conductive pattern formed on an upper surface of the base plate, a second conductive pattern formed on a lower surface of the electrostatic chuck, and a first brazing member joining the first conductive pattern and the second conductive pattern.

6. The substrate fixing device according to claim 1, wherein the base plate includes a first ceramic plate, a second ceramic plate provided on the first ceramic plate, and a second brazing part joining the first ceramic plate and the second ceramic plate, and the second brazing part includes a third conductive pattern formed on an upper surface of the first ceramic plate, a fourth conductive pattern formed on a lower surface of the second ceramic plate, and a second brazing member joining the third conductive pattern and the fourth conductive pattern.

7. The substrate fixing device according to claim 6, wherein the base plate has a cooling path formed by a recessed portion recessed downward from the upper surface of the first ceramic plate and the second ceramic plate provided so as to close an opening of the recessed portion, the third conductive pattern is provided so as to overlap the second ceramic plate in a planar view, and the fourth conductive pattern is provided so as to overlap the third conductive pattern, but not to overlap the recessed portion, in a planar view.

8. The substrate fixing device according to claim 6, wherein the base plate has a first gas flow path which penetrates the base plate in a thickness direction, the electrostatic chuck has a second gas flow path which communicates with the first gas flow path, the first gas flow path has a first hole portion which penetrates the first ceramic plate in the thickness direction, and a second hole portion which penetrates the second ceramic plate in the thickness direction and communicates with the first hole portion, and the second gas flow path is formed so as to penetrate the electrostatic chuck in the thickness direction and to communicate with the second hole portion.

9. The substrate fixing device according to claim 1, further comprising: an electrode built in the electrostatic chuck.

10. The substrate fixing device according to claim 1, wherein the base plate and the electrostatic chuck are made of a ceramic material having the same material as a main component.
